ASSIGNMENT 4
— Build the Frontend for the Bug Report Management —

1. Overview

In Assignment 3, you developed the backend REST APIs for the Bug Report Management System using Node.js, Express.js, MongoDB, Mongoose, and JWT authentication. In this assignment, you will develop a Single Page Application (SPA) using React and Vite to consume the existing Assignment 3 backend APIs.

You must reuse the backend from Assignment 3. Do not create a separate backend or change the API contracts provided in Assignment 3.

2. Learning Outcomes

After completing this assignment, students should be able to:

- Initialize and organize a React project using a maintainable folder structure.
- Consume REST APIs using Axios.
- Manage JWT authentication on the client side.
- Implement protected routes and role-based frontend access.
- Display API data and handle loading, empty, success, and error states.
- Build a responsive interface using Bootstrap or Ant Design.

3. Technologies

Students must use the following technologies:

- React (Vite)
- React Router DOM
- Axios
- Bootstrap **or** Ant Design

Development Tools:

- Visual Studio Code
- Chrome / Microsoft Edge Developer Tools
- Postman

4. Backend Requirements

Reuse the backend application developed in **Assignment 3**. Default Backend URL: **http://localhost:8002**

Required API Endpoints:

Method	Endpoint	Authentication / Authorization
POST	/users/signup	Not required
POST	/users/signin	Not required
GET	/reports	JWT required; returns reports assigned to the authenticated user
POST	/reports	JWT required; Admin role required

For protected requests, send the token in the Authorization header using the Bearer scheme: Authorization: Bearer <token>. The backend remains responsible for enforcing authentication and authorization; frontend route protection alone is not sufficient.

5. Functional Requirements

Task 1. Project Initialization (2.0 points)

Objective: Create a new React project using Vite and organize the source code using a clean and maintainable folder structure.

Requirements:

1. Create the Project

The project name must follow the format:

Assignment4_<memberCode> (ex: assignment4_sangnqce150621)

2. Install Required Packages

Install the following libraries:

- axios
- react-router-dom
- bootstrap or antd

3. Organize the Project Structure:

A recommended structure is shown below:

```
src/
│
├── api/
├── services/
├── pages/
├── components/
├── context/
├── routes/
├── App.jsx
└── main.jsx
```

Students may add additional folders if necessary.

Task 2. User Registration (2.0 points)

Route: /signup

Create a registration page with the fields required by Assignment 3:

- Username
- Email
- Password
- Role (admin, moderator, or user, as defined by Assignment 3)
- Perform basic client-side validation.
- Send the request to POST /users/signup using the request body expected by Assignment 3.
- Display a success message when registration succeeds and a useful error message when it fails.

Task 3. User Authentication (2.0 points)

Route: /signin

- Provide Email and Password fields.
- Send credentials to POST /users/signin.
- After successful login, store the JWT token in localStorage or another appropriate client-side state solution.
- Include the JWT in subsequent protected requests using Authorization: Bearer <token>.
- Redirect the user to an appropriate page after login.
- Display a clear error message if authentication fails.
- Provide a logout function that clears the client-side token and authentication state.
- Note: Clearing the token on the client logs the user out of this frontend. It does not revoke the JWT on the server unless the backend implements token invalidation.

Task 4. Reports Page (2.0 points)

Route: /reports

- Only authenticated users can access this page.
- Retrieve reports using GET /reports and send the JWT in the Authorization header.
- Display only the reports returned by the backend for the currently authenticated user. Do not display reports assigned to other users.
- Display at least Title, Project, Severity, Status, and Assigned To. Use the populated assignedTo information returned by the backend (for example, username).
- Handle loading, error, and empty-data states.
- Handle HTTP 401 responses appropriately, such as clearing invalid authentication state and redirecting to /signin.

Task 5. Admin Create Report Page (2.0 points)

Route: /reports/create

- Create a page that allows an authenticated Admin to create a new bug report.
- Only users with the admin role should be allowed to access this page in the frontend.
- Submit the form to POST /reports using the JWT in the Authorization header.
- The form must include the fields required by Assignment 3: title, description, severity, project, and assignedTo (the ID of the user assigned to handle the report).
- Display success or error messages returned by the backend.
- If a non-Admin user attempts to access the page, block access or display a 403 Forbidden message.
- The backend must still enforce the Admin role; hiding the page in React alone is not sufficient.

- Important: Assignment 3 requires assignedTo to identify an existing user whose role is user. The newly created report receives the default status open from the backend. The frontend must not bypass these backend rules.

6. Submission Requirements

- Submit the complete React project source code. Remove node_modules before submission.
- Include a README with the project name, installation instructions, how to run the frontend, and the backend URL.
- Include screenshots demonstrating the required pages and results, if a screenshot template is provided by the lecturer.
- Submit the project as a .zip or .rar file named assignment4_<memberCode>.zip.

7. Demonstration and Important Rules

- Demonstrate registration, login, and logout.
- Demonstrate the protected Reports page and confirm that it displays only reports assigned to the authenticated user.
- Demonstrate that an Admin can create a report and that a non-Admin cannot use the Admin-only function.
- Show suitable loading, empty, success, and error states where applicable.
- The lecturer may ask students to explain their React components, Axios requests, JWT handling, route protection, and API integration.
- Use the existing Assignment 3 backend. Do not assume an endpoint exists unless it is listed in this specification.
- Total score: 10.0 points (5 tasks × 2.0 points).

